Propagated Presence: A Bluetooth Propagation-Based Method for Automated Classroom Attendance on Mobile Devices

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Abstract

In today's fast-moving, technology-driven world, manual routines are steadily giving way to automated workflows. Yet in many universities, attendance systems remain outdated: students check in once at the start, with no location verification, no continuous tracking, and little defense against cheating. These gaps waste class time, invite errors, and make proxy check-ins easy. To tackle these shortcomings, we propose a solution that is fast, accurate, scalable, and cost-effective: an Android application that turns every student's smartphone into a Bluetooth Low Energy (BLE) beacon. Instead of installing dedicated hardware, our approach reuses the radio already built into modern phones. Each device periodically broadcasts a unique identifier linked to the user, along with session-specific data. At the same time, the lecturer's mobile device broadcasts its own beacon. After each round, every phone uploads its scan log to the server, where a propagation algorithm determines the set of students actually in attendance. Using a round-based cycle—scan, upload, repeat—prevents data gaps and continuously confirms that students remain in the room for the entire session. We also embed real-time validity checks to block impersonation attempts. Because no extra hardware is needed, campus-wide deployment is inexpensive, and the algorithm's design minimizes calibration challenges even when phone models differ. Our experiments show that the method is reliable, highly accurate, and ready to scale. In future work, we plan to adapt the system to other settings, such as workplaces, and to develop a companion iOS app.

Keywords

Bluetooth Low Energy, beacon-based attendance, attendance system, indoor positioning, mobile applications, proximity detection

1. Introduction

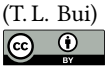
After COVID-19, infection-control rules led schools to switch from slow, hands-on routines to safer digital alternatives—attendance being a prime example [1]. In 2024, investors committed around US$2.4 billion to EdTech, prioritizing classroom-ready tools over high-risk moonshot ideas [2]. Much of this funding is directed toward contactless systems that offer teachers instant attendance statistics and streamline lesson startup [3]. Despite these advances, most teachers still rely on traditional roll call—a time-consuming process prone to errors and student manipulation, as recent smart-attendance pilots have clearly exposed [4].

Although many universities want to replace paper sign-in sheets, standard technological alternatives still face practical limitations. RFID and NFC require students to tap their cards within 10 cm of a reader, creating delays and raising costs due to extra hardware and manual checks [4, 5]. Face recognition is more hands-free but typically only verifies students once at entry, with accuracy degraded by poor lighting, suboptimal camera angles, or masks, and it lacks continuous monitoring [6]. BLE beacon-based solutions leverage students' phones but still depend on fixed in-room beacons and stable Bluetooth environments [4, 7]. In summary, none of these methods fully prevent proxy attendance or work

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seamlessly across rooms without additional system adjustments.

To address these gaps, we propose a smartphone-only approach where each student's device acts as a Bluetooth Low Energy (BLE) beacon using built-in Android and iOS capabilities [7, 8]. Rather than estimating distance from unstable RSSI values, our system accepts all scan events and applies a breadth-first-search propagation algorithm across the dynamically built connection graph. This graph-based decision process is significantly more robust to radio fluctuations and interference typical in crowded classrooms, as it relies on network topology rather than raw signal strength.

The rest of this paper is organized as follows. Section 2 reviews related work and its limitations. Section 3 explains the system architecture. Section 4 describes our propagation algorithm. Section 5 outlines the typical session flow. Section 6 presents the application interface. Section 7 shows our experimental results, and Section 8 concludes the paper.

2. Related Work

Automated attendance methods—ranging from face recognition and geofencing to BLE beacons and QR codes—have shown promise but often depend on constant connectivity, limited trials, and weak liveness or privacy safeguards. This review briefly highlights these gaps to motivate a more robust, offline-capable, and secure attendance solution.

Sharada et al. [9] used Python/OpenCV to log attendance to Google Sheets, but this method needs constant internet, lacks liveness detection, and hasn't been tested on large datasets. P. Sai Vasantha Lakshmi et al. [10] combined geofencing and face recognition to allow check-ins inside a virtual zone without special hardware; however, GPS errors and no liveness or privacy protections limit reliability. Alburaiki et al. [11] used Azure Face API and Mapbox geofencing in a mobile app, but cloud face uploads raise privacy concerns, location accuracy is limited, and evaluation focused on user satisfaction, not objective benchmarks.

Wanjare et al. [12] introduced a contactless attendance system combining face recognition and a dynamic QR code, which refreshes every ten seconds in their Android app. This rolling QR code reduces but does not eliminate replay risks, as students can still share screenshots within the valid window, allowing proxy attendance. The method still faces face recognition's typical issues, such as unreliable performance in low-light and no robust liveness detection. Despite BLE in the title, the paper lacks details on any BLE-based proximity or scanning features. Continuous network connectivity is required to generate QR codes and log attendance, making the system vulnerable if offline or in unstable networks. Kiran Kumar et al. [13] proposed static QR codes embedded in lecture slides for smartphone-based attendance, with backend checks for identity and location. However, the QR codes are static, not time-bound, and the paper does not discuss liveness or secure location verification, making the system susceptible to replay attacks and false check-ins.

Both Hake et al. [7] and Hayati & Nugraha [4] built BLE beacon–based attendance systems but evaluated them only in limited trials without deep technical analysis. Hake et al.'s Android app logs attendance from student BLE broadcasts to a cloud spreadsheet, while Hayati and Nugraha used fixed Cubeacon beacons in classrooms and recorded 91.67–100% "functional success" via an MBaaS backend. However, neither study measures BLE performance in realistic environments—such as radio interference, range spillover, or optimal scanning distance—and both omit analysis of false positive/negative rates and latency with increased class size, raising doubts about scale and reliability across multiple rooms.

Apoorv and Mathur [14] took a different approach by attaching Estimote BLE stickers to student ID cards and cross-checking headcounts with an infrared (IR) people counter. While double-counting can flag blatant mismatches, proxy attendance is still possible if a student hands their ID sticker to someone else. The BLE stickers require yearly battery replacement, introducing extra maintenance, and the paper reports neither BLE range nor RSSI thresholds, so missed scans or readings outside the room remain concerns. Additionally, the IR counter can miscount in dense crowds or blockages, BLE scanning drains phone batteries, and the system does not address offline operation—limiting practical scalability and sustainability for campus-wide or long-term deployments.

Thus, the proposed system addresses several weaknesses in current attendance methods, as it requires no additional hardware and incurs minimal cost. It reliably confirms each student's identity and course enrollment, tracks their actual time in class to prevent proxy check-ins, and continues to run even when the network is offline or unstable. Using a robust signal-propagation algorithm that filters out noise in crowded, obstacle-filled rooms and scales to large classes without confusing devices next door, the system delivers a lightweight, accurate, and highly scalable solution to modern attendance demands.

3. System Architecture

Following Alrowaily [15], our app has student and lecturer roles: students broadcast presence via BLE; lecturers scan and view results. Session management and analytics are handled on the web.

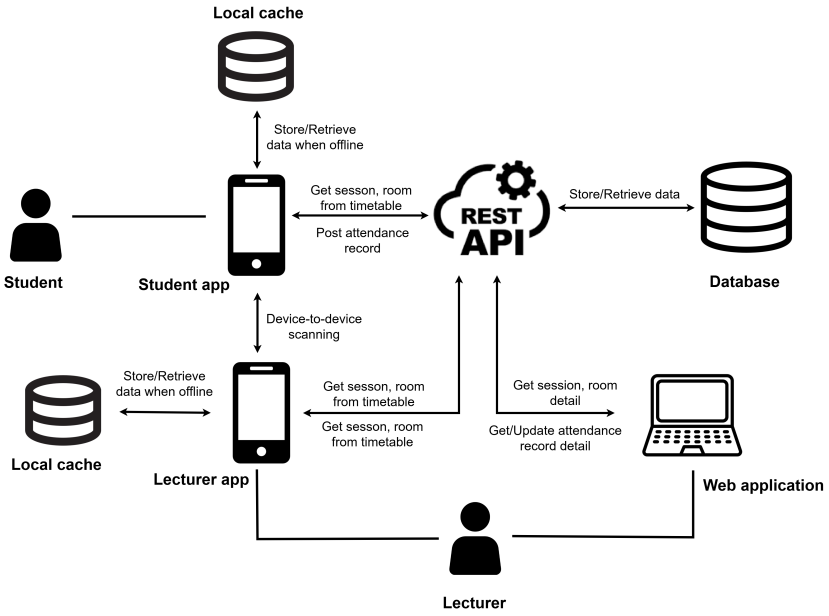


Figure 1: System architecture.

Figure 1 summarizes the system. In both lecturer and student modes, after selecting a session, the app retrieves session parameters via REST, broadcasts BLE identifiers (sessionID, deviceID, roomID), scans for peers, and uploads or caches logs depending on connectivity. The server collects logs, runs the attendance algorithm, and returns statuses for display. The web app manages configuration, processes logs, runs analytics, handles notes and reports, and exposes REST APIs—students see only public notes. All devices broadcast and scan via BLE; scan results sync to the server when online, with local caching for offline use until connectivity returns.

This architecture eliminates extra hardware and enhances privacy by using temporary identifiers. However, network instability and backend load spikes may occur—especially when many devices send events simultaneously. To address this, the app employs an offline-first approach with a secure local cache (e.g., SQLite) to store credentials, session data, and attendance events. Both lecturer and student apps can operate offline: lecturers can sign in, select sessions, and record attendance, while students can check in, with all actions logged locally. When connectivity is restored, the apps automatically upload cached data to the server for verification.

4. Proposed Algorithm

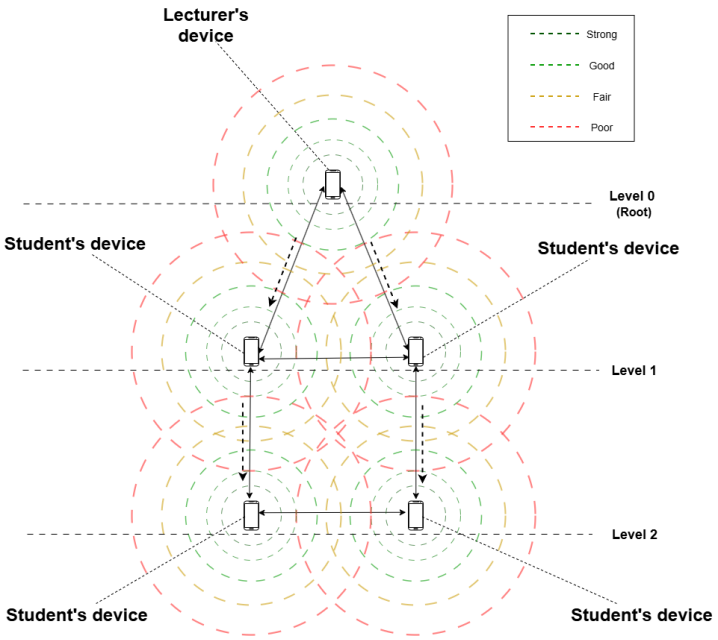


Figure 2: Propagation Flow Illustration

Algorithm 1 Multi-hop BLE Attendance Algorithm

```
function CalculateAttendance(connectivityGraph, lecturerNode)
    attendance ← {lecturerNode}
    queue ← [lecturerNode]
    while queue ≠ ∅ do
        current ← queue.dequeue()
        for neighbor ∈ neighbors(current) do
            if neighbor ∉ attendance then
                attendance.add(neighbor)
                queue.enqueue(neighbor)
            end if
        end for
    end while                                   ▷ Fill-in phase for missed devices
    for device ∈ allDevices do
        if device ∉ attendance and hasAttendedNeighbor(device) then
            attendance.add(device)
        end if
    end for
    return attendance
end function
```

Figure 2 illustrates our multi-hop BLE propagation approach for automated attendance tracking. Each device—whether a lecturer's or student's smartphone—is represented with concentric circles indicating zones of reliable BLE signal strength. Arrows denote the transmission route of the attendance signal: the lecturer's device (root node) initiates a broadcast containing a session identifier. Student devices within the lecturer's direct BLE range receive this broadcast, mark themselves as present, and subsequently rebroadcast the attendance signal. This process continues, allowing the signal to propagate hop-by-hop through intermediary student devices until all present devices are accounted for, even if some are physically distant or out of direct line-of-sight from the lecturer. This peer-to-peer relaying mechanism helps to overcome signal obstructions, dead zones, and extended coverage limitations of BLE.

By treating all student smartphones as potential relays, even students seated far from the lecturer—or behind physical barriers—can participate in automated check-in. Importantly, our system does not establish actual Bluetooth connections between devices; instead, it relies solely on BLE advertising (broadcasting unique identifiers) and scanning modes to collect detection data, which is then used to construct a virtual connectivity graph on the server side. This lightweight approach avoids the overhead of persistent connections while enabling robust propagation analysis. Attendance detection is formalized using a dynamic connectivity graph and a breadth-first search (BFS) traversal scheme, as detailed in Algorithm 1. The process can be broken down into the following key steps: (1) **Connectivity Graph Construction:** Each device scans for nearby devices, recording all detected neighbors and sorting them in descending order of RSSI strength. (2) **Initialization:** The lecturer's device serves as the root node and initiates the session broadcast. (3) **BFS Propagation:** Starting from the lecturer node, BFS traverses the connectivity graph. Each visited device is marked "present" and its neighbors are enqueued for further traversal, ensuring multi-hop reach. (4) **Missed Devices and Fill-in Heuristic:** Devices not reached directly after BFS traversal are considered for fill-in: any unmarked device that can detect (scan) at least one already checked-in device is validated and added to the attendance list, compensating for possible one-way scan misses or undetected paths.

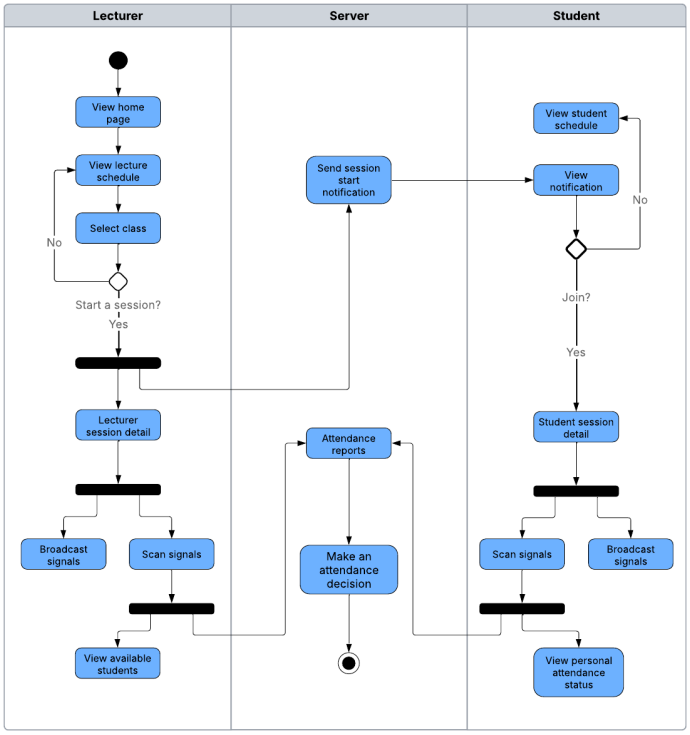


Figure 3: Flowchart of the system

5. Main Flow

When the app starts, it shows the home page. The user first logs in or is identified. Then, the app checks if the user is a lecturer or a student. Figure 3 illustrates the complete workflow for both user roles.

If the user is a lecturer, the app displays their schedule, allowing selection of a class or instant session start. When the session begins, the app notifies students and shows session details. The lecturer's phone then simultaneously broadcasts and scans Bluetooth signals, collecting nearby student devices to record attendance—saving data online or locally for later sync if offline.

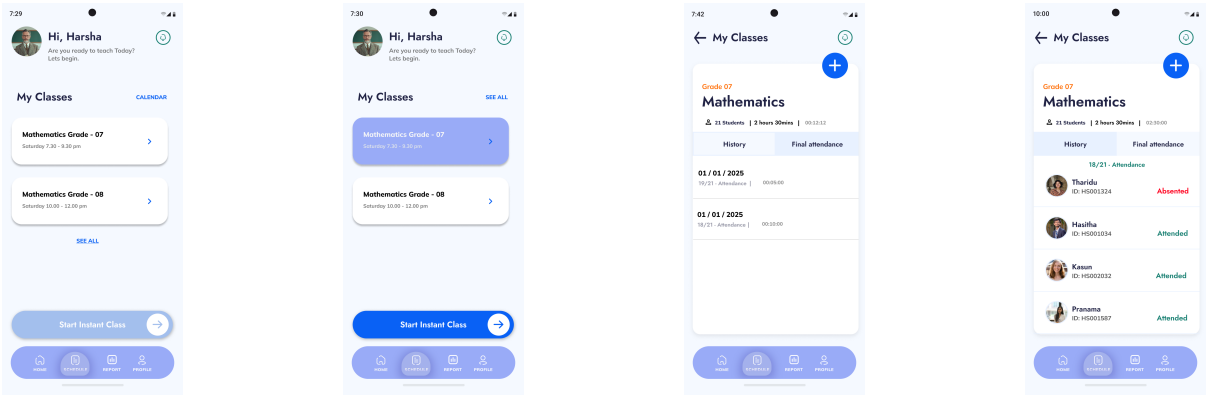
For students, the app displays their schedule. When a session becomes available, students can join it and view session details. Their phone then simultaneously broadcasts and scans Bluetooth signals,

exchanging identifiers with the lecturer and peers. The app records found devices to track attendance, saving the data online or queuing it for later sync if offline.

In both roles, the core steps are: display schedule, start or join a session, show session details, broadcast Bluetooth signals, scan for peers, list detected devices, and record attendance. This simple cycle repeats in rounds (broadcast, scan, record) until the session ends. The same basic flow runs for the lecturer and students, with the lecturer initiating the session and students joining once it starts. The session automatically ends at the scheduled time, when all devices stop BLE broadcasting and the final attendance is calculated.

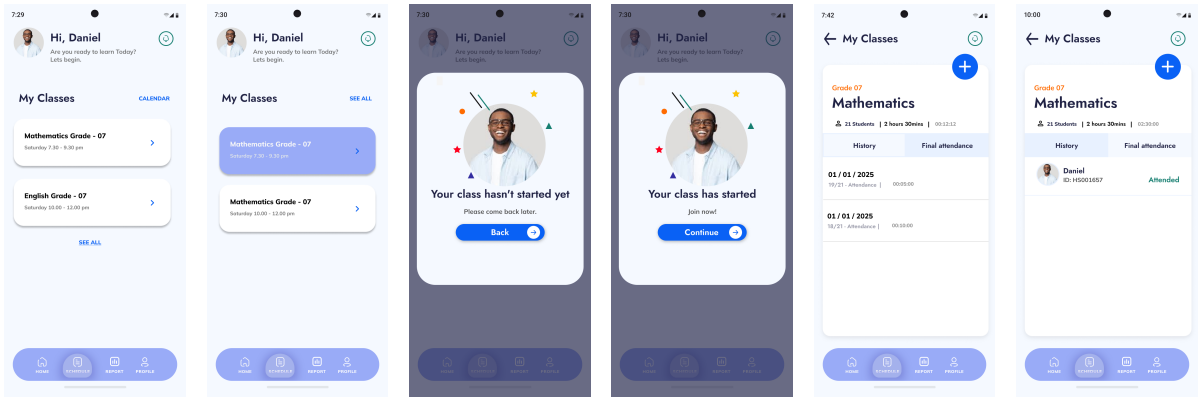
6. Application Interface

As part of the proposed system, the lecturer's UI consists of three core screens that guide the instructor from choosing a class to monitoring attendance rounds. Each screen focuses only on its main function so that the lecturer can work quickly and with minimal distraction.



(a) Schedule view (b) Select a class in schedule view (c) Session detail (d) Final attendance

Figure 4: Lecturer application interface screens.



(a) Schedule view (b) Select class (c) Class hasn't started (d) Join class successfully (e) Session detail (f) Final attendance

Figure 5: Student application interface screens.

Figure 4 presents the lecturer interface workflow. First, a home screen (a) greets the lecturer with a list of upcoming classes shown as clear, selectable cards for fast session startup. After selecting a class (b), the app moves to a detail screen (c) summarizing essential information like student count and session times for confirmation before starting. Once attendance begins, the app (d) monitors in

real time by displaying a session timer, expected student count, and a round-based attendance list that updates automatically as devices are detected. After the session ends, a summary screen (d) provides a scrollable overview of all students, displaying each ID and attendance status for immediate review.

Figure 5a shows the Schedule view, where students see upcoming classes as cards with course names and times. Tapping a card highlights the selection (Figure 5b), confirming the choice. If a class hasn't started, Figure 5c displays a message and "Back" button, prompting the student to return later. When the session goes live, Figure 5d shows a prompt and "Continue" button to join. The Session Detail screen (Figure 5e) lists class details, duration, and attendance rounds labeled "Attended" or "Absent" for tracking presence. Figure 5f gives a summary of total rounds attended, offering a quick overview of the student's attendance.

7. Experimental Results

7.1. Experiment Setup

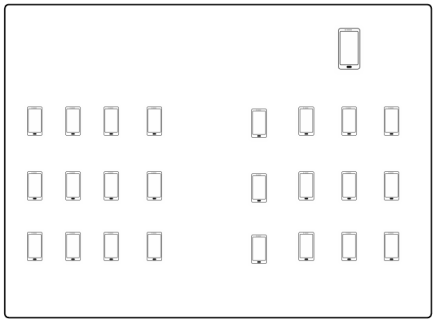


Figure 6: Experimental setup showing the classroom layout and device placement.

Figure 6 describes our live setup. We conducted the test in a standard classroom, with fixed rows of desks and chairs capable of seating around 30 students. For our test, 24 student devices and one lecturer's device ran the application. Student devices were placed on desks or carried in pockets, reflecting typical usage; the lecturer's device remained at the front. The classroom contained common obstacles (tables, chairs, human bodies) and typical interference sources (other mobile devices), matching conditions in prior BLE attendance studies.

The attendance application runs BLE operations automatically on each device. We organize the session into 20 detection rounds of 3 minutes each over one hour. Within each round, the app handles BLE advertising and scanning internally (no user action needed). At the end of each round, the app uploads the detection summary (e.g., which devices were seen) to the server. On the server side, these per-round summaries are aggregated, and presence is determined by counting the number of rounds in which a device was detected and comparing against the threshold. We defined presence as detection in ≥16 out of 20 rounds (3 minutes each). For each scenario, we controlled device availability or placement to simulate attendance patterns, then compared actual detection counts against predicted classifications. In all cases that we conducted, predictions matched results exactly (100% accuracy, no false positives or negatives). Battery drop was 3–4% per device over one hour. Students are marked "present" if detected in at least 80% of rounds (e.g., 16 out of 20), so short absences do not count as missing. The exact number of attended rounds is also recorded for each student. We also note that BLE signals attenuate rapidly when passing through walls or over extended distances. As a result, devices positioned outside the classroom—such as in the hallway or adjacent rooms—receive much weaker and less stable signals, often missing detection in several rounds. In these cases, the attendance threshold is unlikely to be met, effectively reducing false positives for students not physically present inside the room.

7.2. Results

We evaluated the system under representative attendance scenarios, including full presence, temporary absence, late arrival, moderate absence, and extended absence. For each case, we controlled device availability to simulate realistic participation patterns. The system classified a student as "present" if their device was detected in at least 16 out of 20 rounds (80%), and "absent" otherwise. Table 1 summarizes the predicted and observed detection counts for each scenario, along with the resulting classification.

Table 1
Attendance scenarios and results

Scenario	Setup	Pred.	Obs.	Classif.
Full attendance	Present all 20 rounds	20	20	Present
Brief exit	Absent 1–2 rounds (leave/return)	18–19	18–19	Present
Late arrival	Miss first 4 rounds, then present	16	16	Present
Mod. absence	Absent 3–4 arbitrary rounds	16–17	16–17	Present
Ext. absence	Absent ≥5 rounds	≤15	≤15	Absent

7.3. Comparison with Related Work

Continuous monitoring vs. one-time check: Prior work that relies on a single scan at session start cannot detect late arrivals or early departures. Borić et al. demonstrated periodic BLE scans in real lectures to track attendance over time rather than a single initial check [16]. Our method aggregates detection summaries across the entire session, ensuring presence decisions reflect actual time spent in class.

No extra hardware required: Some systems combine BLE scanning with door sensors or other auxiliary devices to reduce false positives (e.g., Borić et al. used door sensors alongside periodic scans [16]). Our algorithm uses only periodic detection summaries sent by the app, maintaining robust performance under typical classroom conditions without additional sensors or infrastructure.

Avoiding calibration overhead: RSSI fingerprinting approaches offer finer-grained positioning but require site surveys and model training before deployment (e.g., Puckdeevongs et al. conducted BLE fingerprinting with neural-network models for classroom attendance [17]). Our detection-count algorithm achieves comparable attendance accuracy without any calibration or offline training, reducing setup time and complexity.

8. Conclusion

This work introduced Propagated Presence, a Bluetooth propagation method for automated classroom attendance that transforms each student's smartphone into a transient beacon and uses a breadth-first-search algorithm to infer true presence without relying on raw RSSI thresholds. By reusing built-in Bluetooth radios and embedding real-time validity checks, the system eliminates the need for dedicated hardware, reducing deployment cost and calibration overhead. The integration of an offline-first architecture and a secure local cache ensures that attendance logging continues uninterrupted even under unstable or absent network connectivity. Queued events are batched and synchronized once connectivity resumes, preserving data integrity and minimizing backend load.

Our live trial with 24 student devices and one instructor device demonstrated flawless detection (100% accuracy) over a one-hour session and negligible battery drain (3–4% per handset), confirming that the graph-based inference and offline-sync design can scale comfortably to typical classroom sizes. These results validate that Propagated Presence delivers a lightweight, reliable, and extensible solution for modern attendance demands, addressing key weaknesses of RFID, NFC, face recognition, and fixed-beacon approaches, particularly by preventing proxy check-ins, tolerating device heterogeneity, and operating smoothly across varied network conditions.

Declaration on Generative AI

The author used generative AI tools for grammar correction and language refinement. All content was reviewed and the author takes full responsibility.

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